

An Exploration and Predictive Analysis of Providence, RI Residential Real Estate Prices

Jack Wright

The data used in the predictive analysis, exploration, and data visualizations was sourced from the Redfin and Walk Score websites. The goals of this report are to build machine learning models that can help predict housing prices in the Providence metro area, and examine the factors that influence housing prices, demand, and economic opportunity. The cost of living crisis has reached every U.S. city, so housing costs and their determinants must be examined. The dataset was examined and cleaned extensively, removing vacant lots, fixing incorrect zip codes, adding in hoa presence, walk score, bike score, and transit score variables. No duplicates were found, and four property types were present, Multi-Family (2-4 unit), Single Family Residential, Condo/Co-op, and Townhouse (distribution shown in Figure A).

After cleaning the data was explored to gain a better understanding of it. A correlation matrix (Figure B) was taken, and beds, baths, and square footage were found to be the most highly correlated with prices at 0.34, 0.57, and 0.61 respectively, however none were very highly correlated. The presence of an hoa was also found to have the largest negative correlation of 0.22. Additionally summary statistics and distribution graphs were analyzed to gain a better understanding of the Providence metro housing market conditions. The median home price in the Providence metro is 514,999, compared to the US median of [410,800](#). Additionally the median number of beds and baths are 4 and 2 respectively. The median square footage is 2,040, and the median walk, bike, and transit scores are 77, 60, and 45 respectively. The most common properties on the market are multi family properties in the 02906 zip code. The three zip codes with the highest price and highest average dollar per square foot were 02903 with 492.25, 02906 with 419.95, and 02919 with 400 dollars per square foot. The average transportation scores of

02903 were 94 for walk, 73 for bike, and 79 for transit. The average transportation scores of 02906 were 80 for walk, 48 for bike, and 70 for transit. The average transportation scores of 02919 were 46 for walk, 34 for bike, and 20 for transit. Zip codes 02903 and 02906 are in denser, desirable areas of Providence with many opportunities and urban amenities nearby. These areas contain a wide variety of housing types, while 02919 is in Johnston and the only on-market home is single family residential.

The goals of this predictive analysis are to provide a framework for the Providence metro area to forecast housing prices in the future, determine economic viability of different neighborhoods as well as infrastructure and economic investment potential, and bring attention to housing affordability as a major issue in the Providence metro area. The variables used in predictive analysis of prices were: number of beds, number of baths, square footage, hoa cost per month, hoa presence, walk score, bike score, transit score, zip code, year built, City, property type, and location. A mix of numeric, categorical, and binary variables were chosen to complete the analysis. Prior to preprocessing I took the log of price due to the high variation in it. I also preprocessed the dataset to fill missing numeric column values with mean values, and for categorical variables to fill with the most common value. Next I created the training and testing split of 70%-30%, and I decided to test an XGBoost, CatBoost, and a Random Forests model. I then ran the models and conducted cross-validation and model comparison (Figure C), where I found the CatBoost model to perform the best with an RMSE of 0.2628, a MAE of 0.1785, and an R-Squared of 0.7058. This model was used for the testing set now, and it performed even better (Figure D), where the RMSE, MAE, and R-Squared were 0.1776, 0.1284, and 0.8909 respectively. The residuals (Figure E) and its distribution (Figure F) were also plotted to visualize the predicted prices compared to the actual. I additionally analyzed the top 10 important features

(Figure G), and found that the top five features of important were the zip code 02906, square footage, the number of baths, the hoa cost per month, and the number of beds. The data exploration and predictive analysis framework can be used to forecast future housing prices in the Providence metro as well as other metro areas.

Image Index

Figure A:

PROPERTY TYPE	
Multi-Family (2-4 Unit)	112
Single Family Residential	104
Condo/Co-op	83
Townhouse	2

Figure B:

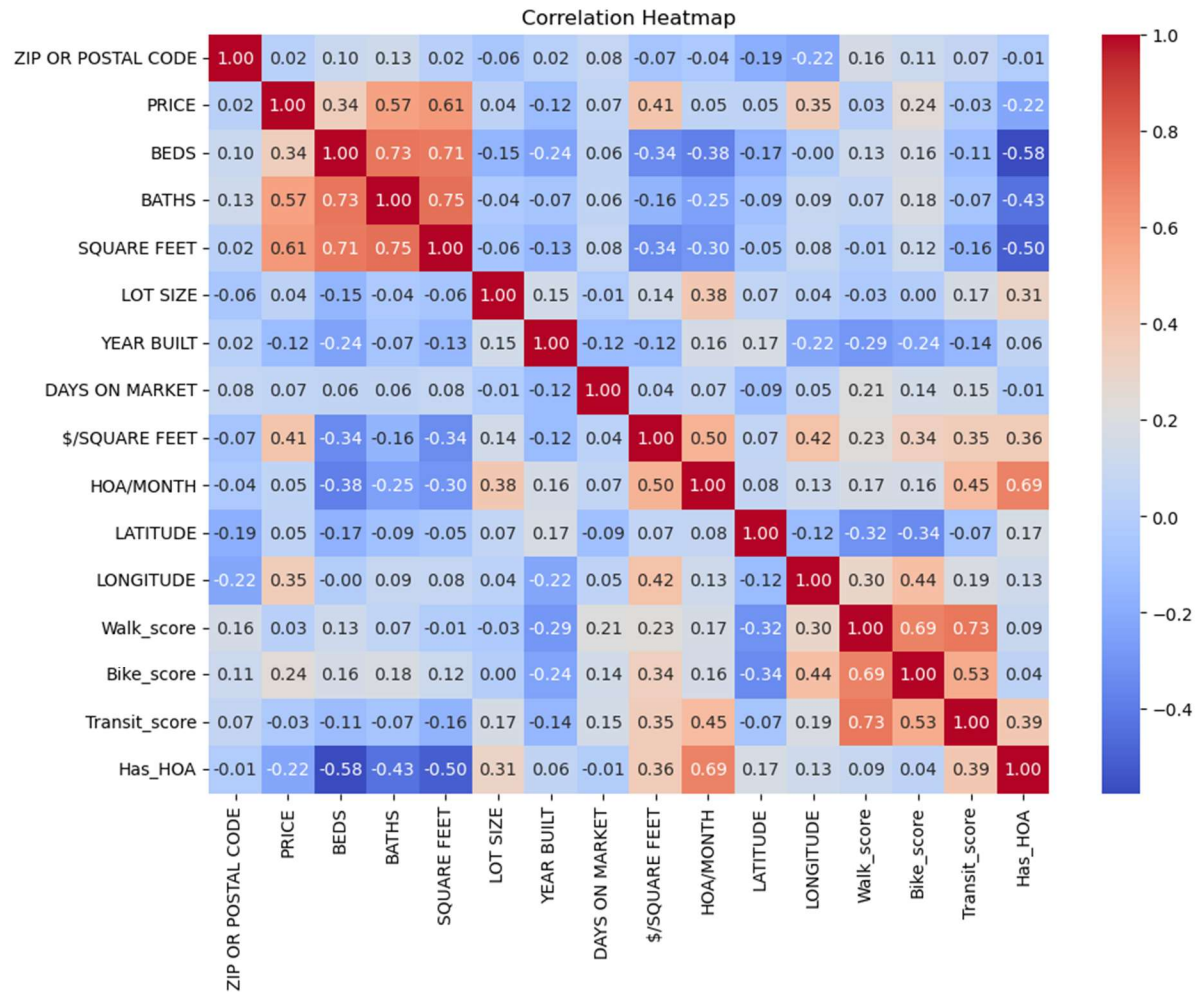


Figure C:

Cross-Validation Results:

	Model	CV RMSE	CV MAE	CV R^2
1	CatBoost	0.262841	0.178460	0.705749
2	RandomForest	0.291601	0.193827	0.639399
0	XGBoost	0.291879	0.199022	0.622118

Figure D:

Final Model Performance on Test Set:

RMSE: 0.1776450769874371

MAE: 0.12843201772145998

R^2 : 0.8909117976970528

Figure E:

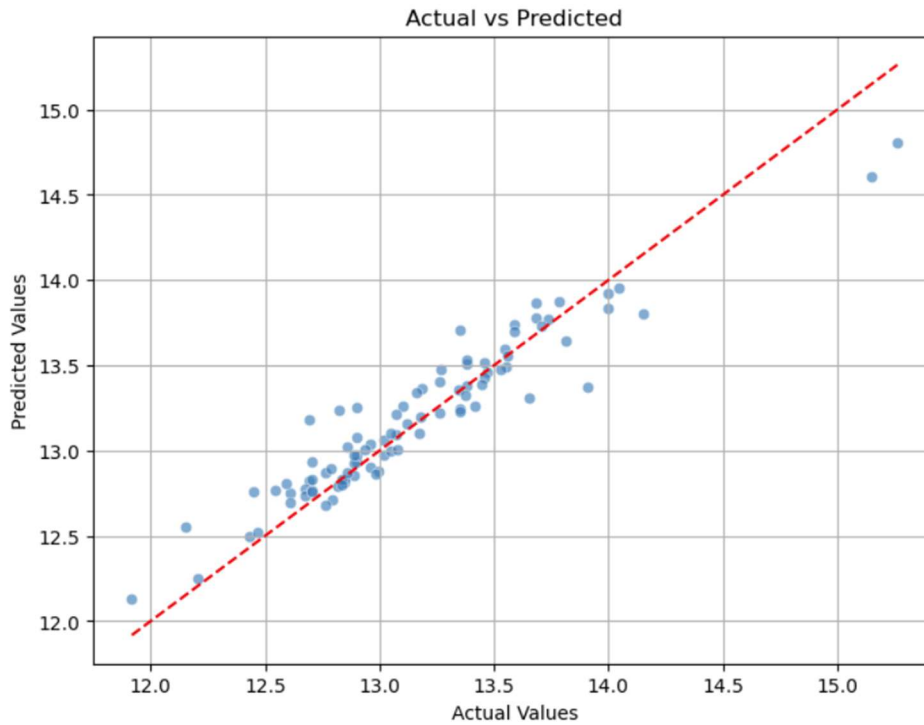


Figure F:

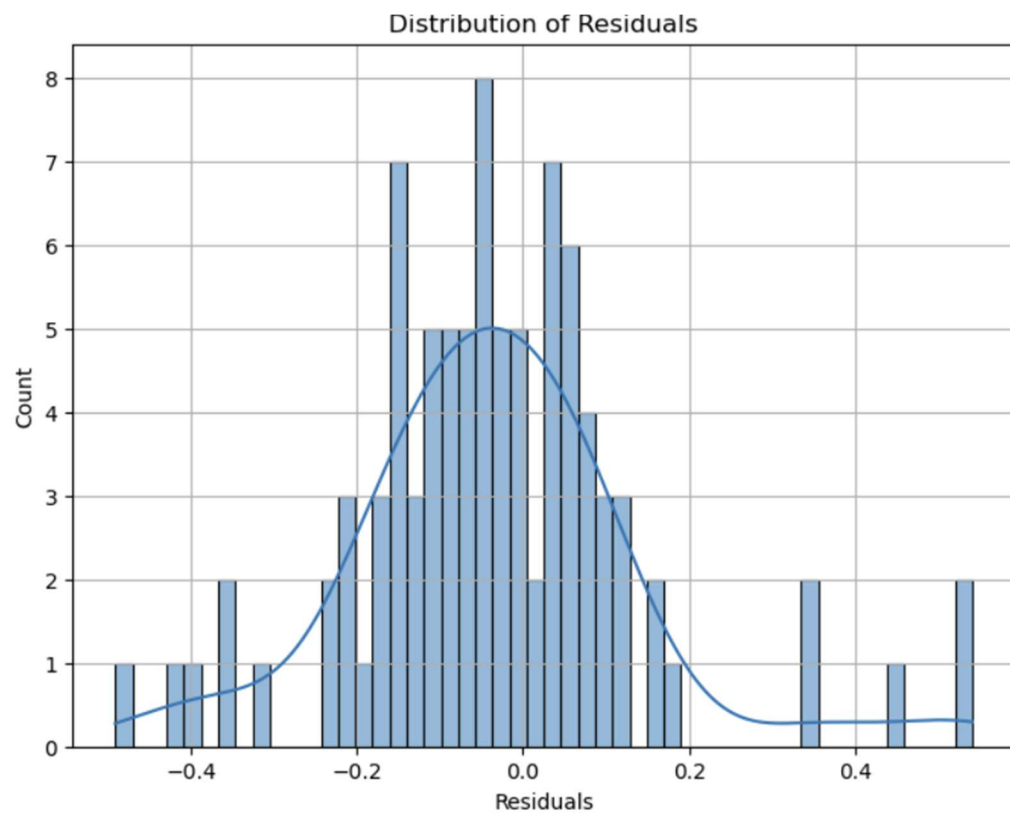


Figure G:

